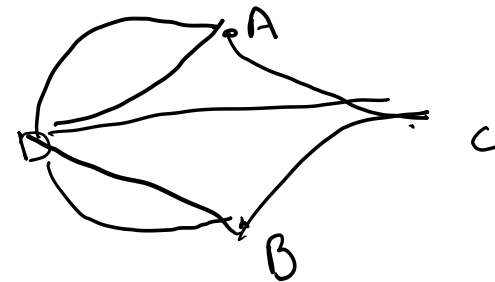


GRAPH THEORY

- Graphs are considered as discrete mathematical structures consisting of vertices and edges connecting vertices.
- Graph theory began in 1736, when Leonhard Euler solved a problem that has been puzzling the citizens of the town of Königsberg in Prussia. The river Pregel divides the town into 4 sections and seven bridges that connected these sections.



Applications

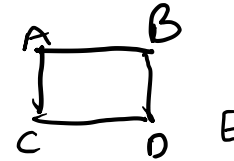
Graph theory is applied in several areas of computer science such as switching theory, logic design, artificial intelligence , formal languages, operating systems, data structures, Software engineering, Image processing, computer graphics and compiler writing.

Graph:

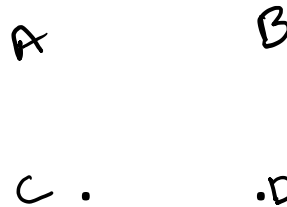
A graph G is a pair of sets (V, E) where “ V ” is a non empty set of vertices (or nodes) or points and “ E ”, a set of edges such that there is a mapping from set E to the set of elements of V (ordered or unordered). If the elements of E are ordered pairs then the graph G is called a directed graph or digraph otherwise undirected graph. If AB is the directed edge, then A is called initial vertex and B is called terminal vertex of AB .



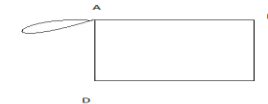
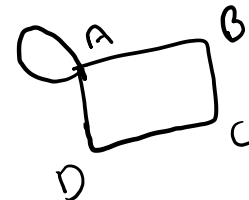
- The pair of nodes that are connected by an edge are called “adjacent nodes”. In the above figure A is adjacent to B and D.
- A node of a graph which is not adjacent to any other node i.e. which is not connected to any other node is called an “isolated node”. In the above figure G is an isolated node.



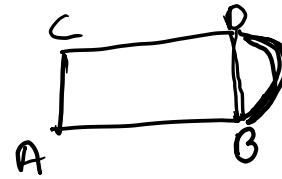
- A graph containing only isolated vertices is called a “null graph”.



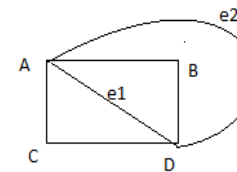
- An edge of a graph which joins a vertex to itself is called a “loop”.



- A graph with a finite vertex set is called a finite graph.
- In a graph, if some pair of vertices are joined by more than one edge, then those edges are called “parallel edges”

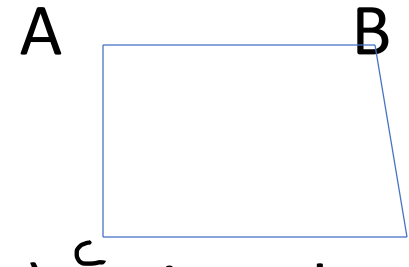


- A graph which contains some parallel edges but no loops is called as “Multigraph”. OR Graphs that have multiple edges connecting the same vertices are called multi graphs

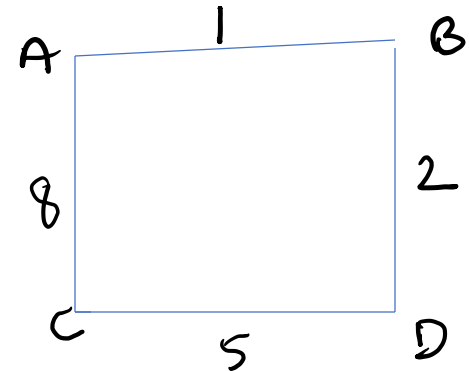


➤ A graph which contains parallel edges or loops or both is called as “General graph”.

➤ A graph without any parallel edges and loops is known as “Simple graph”.



A graph with weights(numbers) assigned to each edge is called a “Weighted Graph



Degree of a vertex

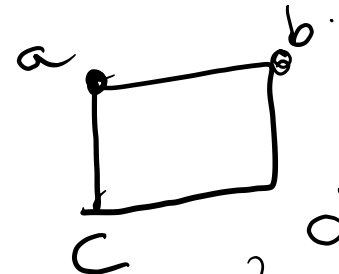
The degree of a vertex in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice the degree of that vertex. the degree of the vertex is denoted by $\deg()$



$$\deg(a) = 2$$

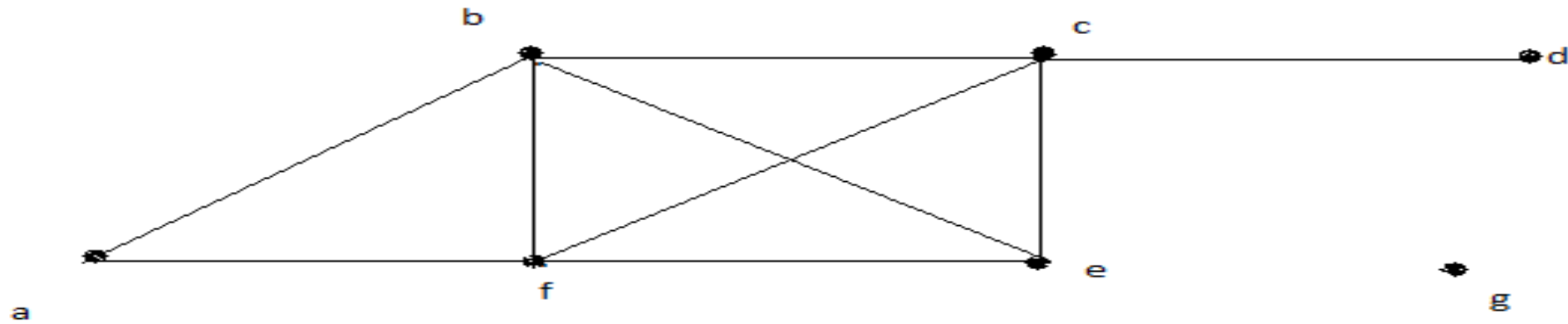
$$\deg(b) = 2$$

$$\deg(c) = \deg(d) = 2$$



Problem

What are the degrees of the vertices in the graph “G” and “H” given below

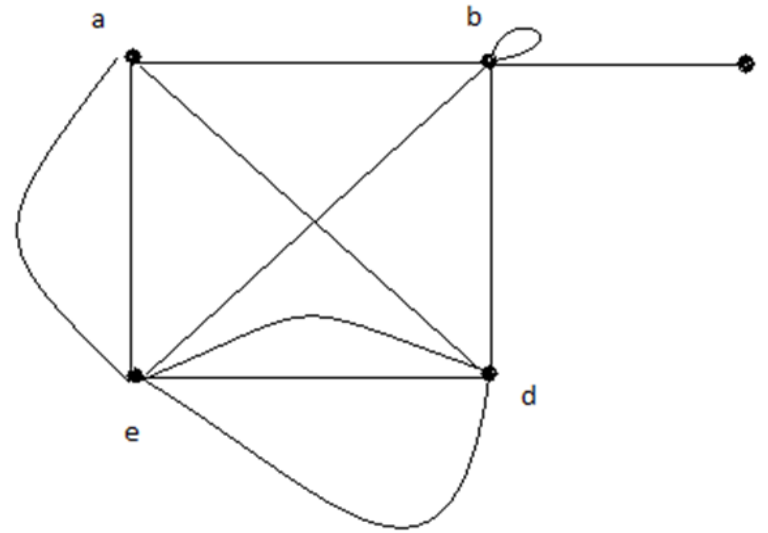


Solution

In “G”

$\deg(a)=2$, $\deg(b)=\deg(c)=\deg(f)=4$

$\deg(d)=1$, $\deg(e)=3$, $\deg(g)=0$



$\deg(a)=4, \deg(b)=\deg(e)=6, \deg(c)=1, \deg(d)=5$
(loop gives degree-2)

Isolated vertex

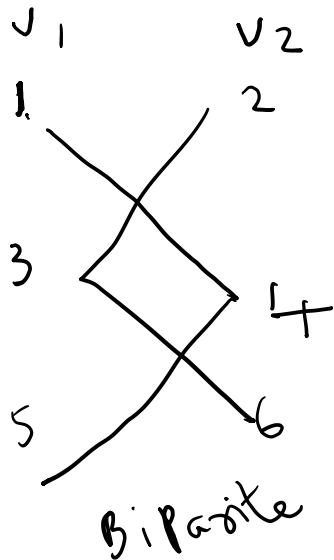
A vertex of degree zero is called isolated vertex.

Pendant Vertex

A vertex is pendant if and only if it has degree one.

Bipartite graph

A simple graph “G” is called bipartite graph if its vertex set “V” can be partitioned into two disjoint sets “ V_1 ” and “ V_2 ” such that every edge in the graph connects a vertex in “ V_1 ” and a vertex in “ V_2 ” (so that no edge in “G” connects either two vertices in “ V_1 ” or two vertices in “ V_2 ”)



Complete bipartite graph

The complete bipartite graph $K_{m,n}$ is the graph that has its vertex set partitioned into two subsets of “m” and “n” vertices, respectively. There is an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.

$K_{3,3}$

$K_{2,3}$

